

CLINICAL INVESTIGATION OF HYPOGLYCEMIC EFFECT OF *EUGENIA JAMBOLANA* IN TYPE-II (NIDDM) DIABETES MELLITUS

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ABSTRACT

The present study was designed to investigate clinically the hypoglycemic effect of *Eugenia jambolana* in type-2 diabetes mellitus. After assaying fasting plasma and urinary glucose, 10 patients of type-2 diabetes mellitus with no previous medication, 10 patients of type-2 diabetes mellitus taking oral hypoglycemic agents with history of inadequate control and six control subjects were given low (2 gm thrice daily) and high (4gm thrice daily) doses of powdered part, aqueous extract and alcoholic extract of seeds of *Eugenia jambolana* for 14 days. On 15th day blood and urine samples for glucose were taken. Based on results obtained it was found that *Eugenia jambolana* has significant hypoglycemic activity both in low and high doses

INTRODUCTION

The jambul fruit is a well known fruit, the tree of which has been cultivated in the Indo-Malaysian region for a long time. It is considered to be native of Indo-Pak and further east. It grows abundantly in rainy season. It has two varieties. The big one is oval in shape and is commonly called as Sava-jamun. The small one is round in shape and is commonly called as Kutta-jamun. The bigger variety is sweeter than the smaller one. The fruit is a juicy berry with a single seed. It is black outside and violet inside. It has a sourish sweet pulp and a greenish yellow seed. The jambul fruit contains minerals (Calcium, Phosphorus, Iron) and vitamins (vitamin C and B. complex) and a very small amount of proteins and fat. Calories value is 62/100 gm (edible portion). Traditionally *Eugenia jambolana* is used for various diseases e.g., Diabetes mellitus, diarrhoea, piles, secretagogue, and to stimulate liver functions (Bakhru, 2000).

In the present investigation, the effect of powdered, aqueous and alcoholic extracts of

seeds of *Eugenia jambolana* were investigated clinically in type-2 diabetic patients.

MATERIAL AND METHODS

The present study was carried out in the laboratories of Hamdard Institute of Pharmaceutical Sciences, Islamabad and Army Medical College, Rawalpindi. The following criteria was used to include or exclude the patients in the study.

Inclusion Criteria

1. Type 2 diabetic patients with fasting plasma glucose level equal to or greater than 140 mg/dl of blood (WHO study group on diabetes mellitus, 1985, Geneva technical report series 727) (WHO 1985) with out any detectable/visible complications.
2. Type 2 diabetic patients taking oral hypoglycaemic agents with history of inadequate control of blood glucose with these agents.
3. Normal healthy subjects with no family history of diabetes mellitus.

4. The patients and control subjects were of either sex (male or female) between the ages of 35-60 years.

Exclusion Criteria

1. Pregnant or nursing patients.
2. Smokers
3. Patients with GIT, hepatic, cardiovascular, renal or endocrine disorder (other than diabetes mellitus) which can interfere with the absorption, metabolism and excretion of the study plant.
4. Patients with any complication of diabetes mellitus.
5. Patients suffering from type-1 (IDDM) diabetes mellitus.

Subjects

The selected subjects (patients and controls) were medically examined and given code numbers and were asked to present themselves on a specified date for sample collection. They were requested to come with fasting (no food before 12 hours) and to void their morning urine and to drink 250ml glass of water before coming for testing (Bahajiri *et al.*, 2000). Patients already taking oral hypoglycaemic agents were requested to take their usual medicine and food after sampling.

Blood Sample

Blood samples (3-5 ml) were drawn from each patient and control subject by venepuncture through plastic disposable syringes. The blood samples were collected in clean oven dried glass bottles which were previously rinsed with 1% sodium fluoride, 3% potassium oxalate solution to prevent coagulation and glycolysis. The plasma was separated after centrifugation. Any sample showing haemolysis was discarded. After separation of plasma, it was transferred to clean, previously acid rinsed, washed and oven dried glass bottles with plastic caps. The plasma glucose estimation was done immediately on the same day by kit method (Trinder, 1969).

Plant

Seeds of *Eugenia jambolana* were purchased from the local market. Dr. Mir Ajab Khan department of biological sciences Quaid-

i-Azam University Islamabad identified the seeds of the plant.

The seeds were shade dried, pulverised by a mechanical grinder and passed through 40-mesh sieve.

Aqueous Extract

After grinding in an electric grinder, the powder was soaked in equal amount of water and stirred intermittently and then left overnight. The macerated pulp was then filtered through a coarse sieve and the filtrate was dried at reduced temperature. This dry mass served as aqueous extract of the plant for experimentation (Vats *et al.*, 2002).

Alcoholic Extract

Alcoholic extract was prepared by powdering 1 kg plant material in an electric grinder. The powder was then mixed with 500 ml of alcohol and kept at room temperature for 36 h. The slurry was stirred intermittently for 2 h and left overnight. The mixture was then filtered and the filtrate was freed from solvent under partial vacuum (71 mmHg) at 35 – 45°C to yield pulp. A few drops of silicon emulsion were added near the end of distillation to avoid frothing. The final residue collected was a thick paste. This was dried at reduced temperature. This dried mass served as alcoholic extract for experimentation (Vats *et al.*, 2002).

General Plan of Study

The fasting blood and urine samples from patients and control subjects were assayed for respective glucose levels.

10 patients of type 2 diabetes mellitus and 10 type-2 diabetic patients taking oral hypoglycaemic agents (with history of inadequate control) were given dry powdered seeds of *Eugenia jambolana* for fourteen days. On 15th day blood and urine samples of glucose were taken.

After an interval of one-week fasting blood and urine samples were again taken from these patients.

Table

Table showing effects of dry powdered part, aqueous extract and alcoholic extract of seeds of *Engenia jambolana* on blood and urinary glucose in type-2 diabetic patients with no previous medication, taking oral hypoglycemic agents with history of inadequate control and control subjects before and two weeks after administration. The values are given as Mean $\pm$ SD (n=10+10+6)

	Blood glucose (mg/dl)		Urinary Glucose	
	Before Administration	After Administration	Before Administration	After Administration
<i>Dry Powdered Part</i>				
No previous medication				
Low dose	194 $\pm$ 8.51	133.6 $\pm$ 6.54*	Glycosuria	No. Glycosuria
High dose	206.6 $\pm$ 21.03	103.8 $\pm$ 9.33*	Glycosuria	No. Glycosuria
Taking oral hypoglycemics				
Low dose	158.4 $\pm$ 10.50	130.6 $\pm$ 7.89*	No. Glycosuria	No. Glycosuria
High dose	162.8 $\pm$ 9.31	105 $\pm$ 14.57*	No. Glycosuria	No. Glycosuria
Control				
Low dose	70	80 ^{NS}	No. Glycosuria	No. Glycosuria
High dose	90	70 ^{NS}	No. Glycosuria	No. Glycosuria
<i>Aqueous Extract</i>				
No previous medication				
Low dose	189.8 $\pm$ 7.49	134.2 $\pm$ 3.19*	Glycosuria	No. Glycosuria
High dose	200.4 $\pm$ 7.30	115.2 $\pm$ 16.54*	Glycosuria	No. Glycosuria
Taking oral hypoglycemics				
Low dose	195 $\pm$ 12.74	135.6 $\pm$ 14.11*	Glycosuria	No. Glycosuria
High dose	216 $\pm$ 15.57	101 $\pm$ 9.61*	No. Glycosuria	No. Glycosuria
Control				
Low dose	85	90 ^{NS}	No. Glycosuria	No. Glycosuria
High dose	70	65 ^{NS}	No. Glycosuria	No. Glycosuria
<i>Alcoholic Extract</i>				
No previous medication				
Low dose	209 $\pm$ 26.31	122 $\pm$ 10.36*	Glycosuria	No. Glycosuria
High dose	226 $\pm$ 17.10	98 $\pm$ 11.51*	No. Glycosuria	No. Glycosuria
Taking oral hypoglycemics				
Low dose	163 $\pm$ 10.36	115 $\pm$ 13.69*	No. Glycosuria	No. Glycosuria
High dose	153 $\pm$ 9.87	88 $\pm$ 11.51*	No. Glycosuria	No. Glycosuria
Control				
Low dose	65	85 ^{NS}	No. Glycosuria	No. Glycosuria
High dose	100	80 ^{NS}	No. Glycosuria	No. Glycosuria

Low dose: 2 g tid; High dose: 4 g tid; * Significant; NS: Non significant

These patients were given aqueous extract of *Eugenia jambolana* seeds for 14 days. On 15th day blood and urinary samples of glucose were taken. After an interval of one-week fasting blood and urine samples for the monitoring of glucose level were again taken from these patients. These patients were given alcoholic extract of the *Eugenia jambolana* seeds for 14 days. On 15th day blood and urinary samples of glucose were taken.

Out of the ten patients five received low (2 g thrice daily) dose and five received high dose (4 g thrice daily) of the plant.

Six healthy subject were kept as control three subject received low dose and three subject received high dose of powdered, aqueous and alcoholic extracts of seeds of *Eugenia jambolana* as described above.

All the patients and control subjects were monitored for any adverse reaction of the plant.

Plasma assay of glucose was done by kit method and urinary glucose was estimated by strip method (Burtis and Ashwood, (1998). The data was statistically evaluated (Glazer, 2001).

RESULT

The results are summarized in the table and graph.

Untoward Effects

Mild headache initially was complained by some patients belonging to all the groups. The basis of headache was psychological.

DISCUSSION

Diabetes mellitus has been defined as a syndrome of abnormal carbohydrate

metabolism, resulting in hyperglycaemia with acute metabolic complications and chronic vascular, neurogenic and orthopedic complications affecting many organs of the body (Jamil *et al.*, 2001).

A significant decrease was noted in mean concentration of plasma glucose two weeks after administration of both low (2gm tid) and high (4gm tid) dose of powdered part, aqueous extract and alcoholic extract of seeds of *Eugenia jambolana* to type-2 diabetic patients. There was a marked fall in mean concentration of plasma glucose with high dose of seeds of *Eugenia jambolana* as compared to that of low dose. The mean value of plasma glucose was in the normal controlled range of diabetes mellitus (although at upper limit of normal with low dose).

Glycosuria disappeared two weeks after administration of both low and high dose of seeds of *Eugenia jambolana*.

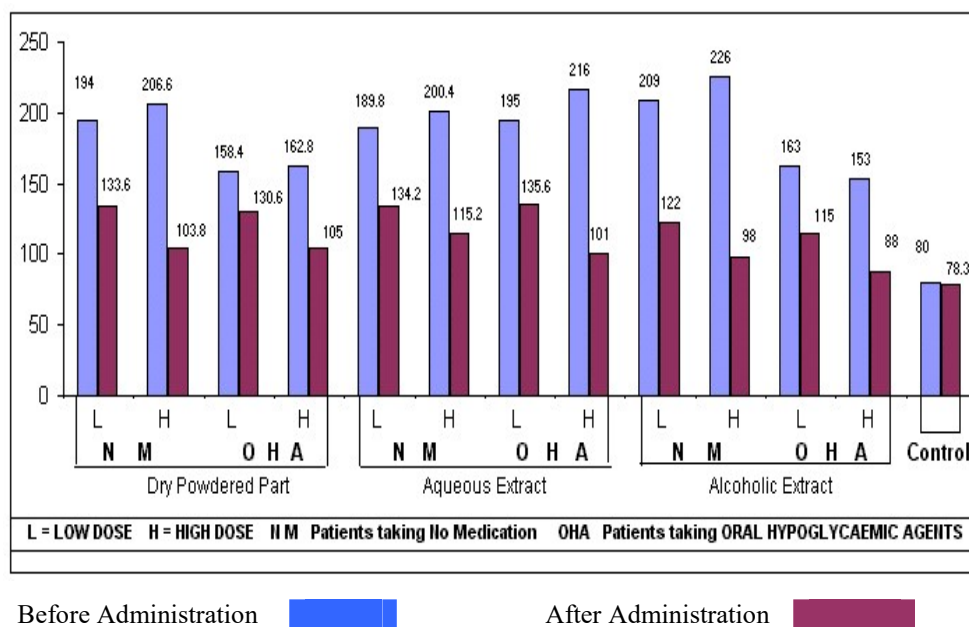


Fig. 1: comparison of effect on blood glucose in type – 2 diabetic patients and control subjects before and two weeks after administration of dry powder part, aqueous and alcoholic extract of fruits of *Eugenia jambolana*

These results confirm the claims by Bakhru, (2000) that:

- Jambul fruit is regarded in traditional medicine as a specific against diabetes mellitus because of its effect on the pancreas. The seeds in doses of 3 g given with water reduces the quantity of sugar in urine. In Aurveda the dried burnt and powdered inner bark of Jambul tree given in a dose of 65-135 mg twice daily reduces the blood and urinary glucose.
- But in another study when streptozocin diabetic rats were treated for 17 days with a decoction of *Eugenia jambolana* (Myrtaceae) leaves (15% W/V) as a substitute for water, the *Eugenia jambolana* leaf decoction was found to have no antidiabetic activity (Pepato et al., 2001).
- In one more study treatment with 400 mg/day of aqueous extract of *Momordica charantia* and *Eugenia jambolana* for 15 days substantially prevented hyperglycaemia and hyperinsulinemia induced by a diet high in fructose (Vikrant et al., 2001).

Mechanism of Action

The seeds of the jambul fruit contain a glycoside, jambolin, which is believed to have the power to check the pathological conversion of starch into sugar in cases of increased production of glucose (Bakhru, 2000).

CONCLUSION

Eugenia jambolana seeds have significant blood glucose lowering effect and can be given to type-II diabetic patients with no previous medication alone or in combination to patients taking oral hypoglycemic agents with inadequate control of blood glucose both in low or high doses depending upon their plasma glucose concentration.

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